

A DSP program uses `floor()` instead of `round()` during quantization. For the samples $[1.78, 2.62, -1.37, -2.81]$, identify the incorrect outputs and debug the code.

~ Presented by:

S. pooja [192412450]

INTRODUCTION

- Quantization converts continuous signal amplitudes into discrete integer values.
- The `round()` function gives the nearest integer, while `floor()` always rounds down.
- Using `floor()` instead of `round()` produces incorrect quantized outputs.
- This problem identifies the incorrect outputs and debugs the program by replacing `floor()` with `round()`.

OUTPUT

```
1  clc;
2  clear;
3
4  % Input samples
5  x = [1.78 2.62 -1.37 -2.81];
6
7  % Incorrect quantization
8  y1 = floor(x);
9
10 % Correct quantization (Debugged)
11 y2 = round(x);
12
13 disp('Input Samples:');
14 disp(x);
15
16 disp('Output using floor() (Incorrect):');
17 disp(y1);
18
19
```

Input Samples:

1.7800	2.6200	-1.3700	-2.8100
--------	--------	---------	---------

Output using floor() (Incorrect):

1	2	-2	-3
---	---	----	----

Output using round() (Correct):

2	3	-1	-3
---	---	----	----

Incorrect Outputs:

1.78 : floor = 1, round = 2
2.62 : floor = 2, round = 3
-1.37 : floor = -2, round = -1



```
clc;
clear;
% Input samples
x = [1.78 2.62 -1.37 -2.81];
% Incorrect quantization
y1 = floor(x);
% Correct quantization (Debugged)
y2 = round(x);
disp('Input Samples:');
disp(x);
disp('Output using floor() (Incorrect):');
disp(y1);
disp('Output using round() (Correct):');
disp(y2);
fprintf('\nIncorrect Outputs:\n');
for i = 1:length(x)
    if y1(i) ~= y2(i)
        fprintf('%0.2f: floor = %d, round = %d\n', x(i), y1(i),
y2(i));
    end
end
end
```

THANK YOU